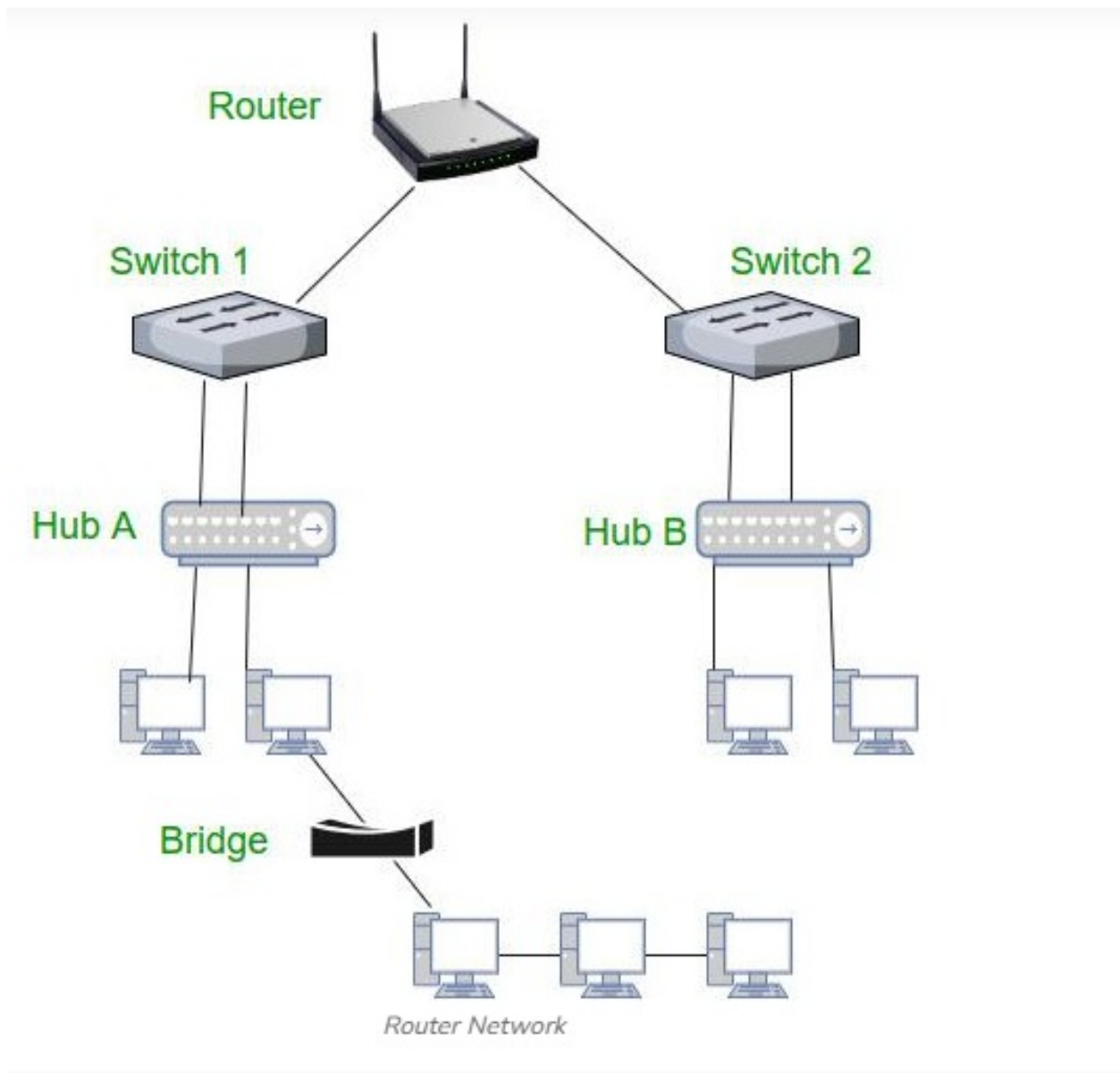


NETWORK FUNDAMENTALS & ROUTER BASICS

14-Section Beginner Notes • Visual Network Structure • Cisco Router Commands



1. Prelude

- Computer networking connects devices and services so information can move between endpoints.
- Learning path: devices → media → addressing → protocols → switching/routing → configuration → troubleshooting.

2. Introduction to Network

- A network is a group of connected devices that communicate and share data or resources.
- PAN = Personal Area Network; LAN = Local Area Network; MAN = Metropolitan Area Network; WAN = Wide Area Network.
- Communication requires a sender, receiver, medium, addressing and protocols.

3. Networking Models

- OSI has seven layers: Physical, Data Link, Network, Transport, Session, Presentation and Application.
- TCP/IP groups functions into Network Access, Internet, Transport and Application.
- Models make networking easier to understand and troubleshoot.

4. Cables

- UTP/twisted-pair Ethernet is common for LAN connections.
- Fiber optic uses light and is useful for high bandwidth and long distances.
- Coaxial uses shielded copper and is common in cable-TV and some legacy applications.
- Choose cable according to interface, distance and bandwidth.

5. Network Topologies

- Star: endpoints connect to a central switch.
- Bus: shared backbone; mostly historical for Ethernet.
- Ring: devices form a ring.
- Mesh: multiple links provide redundancy.
- Hybrid: combination of topology types.

6. IP Addressing

- IPv4 uses 32-bit addresses such as 192.168.1.10.
- A subnet mask separates network and host portions.
- Default gateway is normally the router interface used to reach other networks.
- Private ranges: 10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16.
- IPv6 uses 128-bit addresses.

7. Protocols and Utilities

- Ethernet — LAN frames.
- ARP — IPv4-to-MAC resolution.

- IP — logical addressing and routing.
- TCP — reliable transport.
- UDP — connectionless transport.
- DNS — name resolution.
- DHCP — automatic IP configuration.
- HTTP/HTTPS — web communication.
- SSH — secure remote administration.
- ping — reachability testing.
- traceroute/tracert — path discovery.

8. Introduction to Router

- A router connects different IP networks and forwards packets using a routing table.
- Router interfaces connect the device to LANs, WANs or other networks.
- A router interface can serve as a default gateway.

9. Router Basics

- Components: CPU, RAM, NVRAM, Flash, interfaces and console port.
- Boot process: POST → bootstrap → locate IOS → load IOS → load startup configuration.
- Cisco IOS modes include User EXEC, Privileged EXEC and configuration modes.

10. Network Devices

- Switch — mainly Layer 2; forwards Ethernet frames using MAC information.
- Hub — Layer 1 repeater; largely obsolete in modern Ethernet.
- Router — Layer 3 device; connects IP networks.
- Server — provides services such as web, DNS, DHCP, files or applications.

11. Router Connections and Configuration

- Connect the host or switch to the correct router interface.
- Assign an IP address and subnet mask.
- Use no shutdown when required.
- Set the host default gateway to the router interface.
- Verify using show ip interface brief and ping.

12. Router Security and Remote Access

- Use strong credentials.
- Prefer SSH over Telnet for encrypted remote management.
- Restrict management access to trusted users/networks.
- Never publish real passwords, private keys or confidential configurations.

- Use a suitable MOTD banner where required.

13. Cisco Packet Tracer Practice

- Start with PC → Switch → Router.
- Configure IPv4 addressing and the router gateway.
- Verify using show commands.
- Ping the gateway first, then test other networks.
- Build a two-LAN router lab after the first LAN works.

14. Conclusion and Quick Revision

- Switch = connects devices inside a LAN.
- Router = connects different IP networks.
- Hub = basic Layer 1 repeater.
- Server = provides network services.
- DNS = name to IP.
- DHCP = automatic IP configuration.
- SSH = secure remote CLI.
- Ping = reachability test.

Router Commands — Quick Reference

This section is added specifically for your router practice. Commands are grouped by what you normally do in Cisco IOS.

1. Enter and exit modes

```
Router> enable
Router# disable
Router# configure terminal
Router(config)# end
Router(config)# exit
```

2. Basic identification and configuration

hostname R1 — Set the router hostname.

show running-config — Display the current running configuration.

show startup-config — Display the saved startup configuration.

show version — Display IOS, hardware and version information.

show clock — Display the device clock.

show history — Display previously entered commands.

3. Interface configuration

```
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# description LAN_CONNECTION
Router(config-if)# exit
```

- **ip address** assigns an IPv4 address and subnet mask.
- **no shutdown** administratively enables the interface.
- **description** adds a useful label to the interface.

4. Verification commands

show ip interface brief — Quick view of interface, IP address and status.

show interfaces — Detailed interface information and counters.

show ip route — Display the IPv4 routing table.

show arp — Display the ARP table.

show cdp neighbors — Display directly connected Cisco neighbors when CDP is enabled.

ping 192.168.1.10 — Test IP reachability.

traceroute 192.168.1.10 — Trace the path to a destination.

5. Save configuration

```
Router# copy running-config startup-config
```

This saves the current running configuration so it can be restored after a reload.

6. Basic security commands

```
Router(config)# enable secret
Router(config)# username admin secret
Router(config)# line console 0
Router(config-line)# login local
```

```
Router(config-line)# exit
```

Use strong passwords in real environments. Never place real credentials in public notes or GitHub.

7. SSH remote access — basic lab example

```
Router(config)# hostname R1
Router(config)# ip domain-name example.local
Router(config)# username admin secret
Router(config)# crypto key generate rsa
Router(config)# line vty 0 4
Router(config-line)# login local
Router(config-line)# transport input ssh
```

The exact RSA/SSH prompts can vary with Cisco IOS and Packet Tracer versions.

8. Banner command

```
Router(config)# banner motd #Authorized access only.#
```

Displays a Message of the Day banner when users connect.

9. Troubleshooting command order

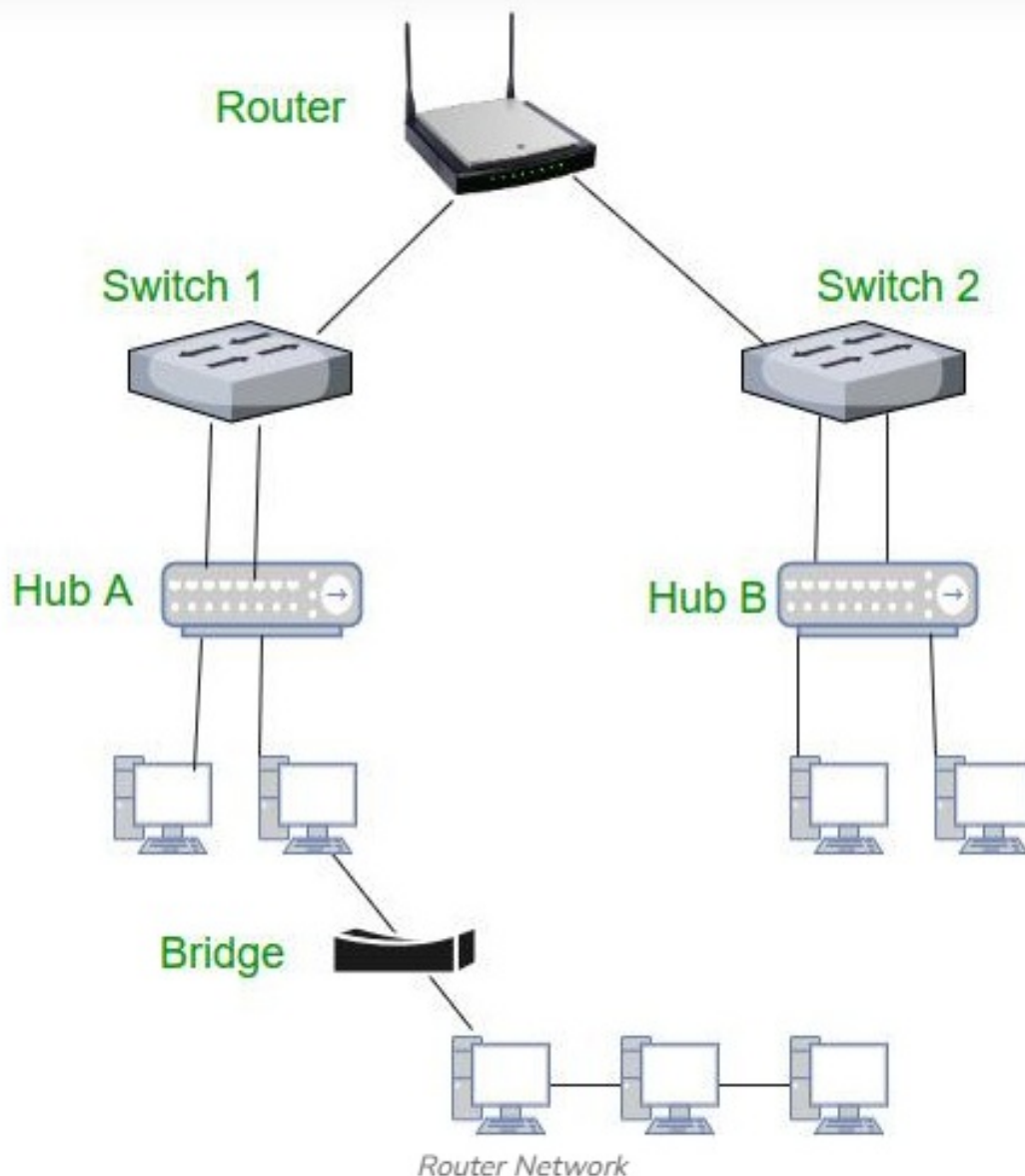
- show ip interface brief — check interface status and IP.
- ping the local gateway — test local connectivity.
- show ip route — check known networks/routes.
- show interfaces — inspect errors and link information.
- traceroute — investigate the path to a remote destination.

10. Mini practice

```
Router> enable
Router# configure terminal
Router(config)# hostname R1
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# end
Router# show ip interface brief
Router# copy running-config startup-config
```

Network Structure — Visual Reference

Your supplied topology is included here as the main structure reference.



Read it from the bottom upward: end devices connect to hubs/switches, switches connect toward the router, and the router connects different networks.

Real Hardware Images — Final Appendix

The final appendix is reserved for real photographs of a router, switch, hub, server, Ethernet cables and connectors. Exact photographs can be inserted when you provide them.